

SOLVING CHEMICAL ENGINEERING PROBLEMS WITH COMSOL

Major Project report

Submitted in Partial Fulfilment of the Requirements for the Degree of

BACHELOR OF TECHNOLOGY

in

CHEMICAL ENGINEERING

by

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DECLARATION

I hereby declare that the Report of the Major Project on ‘Solving Chemical Engineering Problems With COMSOL’ which is being submitted to National Institute of Technology Karnataka, Surathkal, in the partial fulfilment of the requirements for award of the Degree of Bachelor of Technology, Department of Chemical Engineering, is a Bonafede report of the work carried out by me. The material in the report has not been submitted to any university or Institute for the award of any degree.

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CERTIFICATE

This is to certify that the Major Project Report entitled “**Solving Chemical Engineering Problems With COMSOL**” submitted by **H S Manoj Gowda (171CH018), Vinay Mane (171CH052), Sourav Jain (171CH055)** as a record of the Major Project performed by them is accepted as the Major Project Report Submission in partial fulfilment of the requirements for the award of **Bachelor of Technology in Chemical Engineering** in the Department of Chemical Engineering, National Institute of Technology Karnataka, Surathkal.

Course Coordinator

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ABSTRACT

This thesis seeks to create a high-fidelity model of the Multiphysics present in a typical monolith catalytic converter converting carbon monoxide to carbon dioxide. The system is fully described by energy, momentum and mass equations, all of which are coupled. Combustion was modelled as a surface reaction occurring on the catalyst walls, with rate model based on Arrhenius assumptions. Furthermore, Darcy's law was applied to describe flow of gases in porous medium

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1. INTRODUCTION

1.1 SIMULATION

In the past decade the importance of simulation has been growing as the computers got faster and better. Simulation has been a really helpful tool building models as opposed to real life construction, and the economical expenses associated with it. Its best feature is the ease with which we can switch and modify parameters to get a new analysis. Sometimes as the complexity increases the expenses of computing the solutions also increase but it's better than the alternative, since it would involve a variety of parameters all interlinked to one another.

Simulation is an art. It's not always the best approach to solve one's problems. Very often people become too focused and engrossed with modelling every little detail imaginable, creating complexities. Simulations' need is to ease the process, not make it harder to grasp. To master this art one must understand and be able to figure out the necessary details from the unnecessary ones. Being able to optimize a few key parameters is more meaningful than doing the same with a dozen different variables.

There are multiple numerical solvers' packages available commercially to solve the partial equations that govern the model with accuracy and stability, and the one we're going to use for this project is COMSOL Multiphysics.

1.2 COMSOL

COMSOL Multiphysics is a cross-platform finite element analysis, solver and Multiphysics simulation software. It allows conventional physics-based user interfaces and coupled systems of partial differential equations (PDEs). COMSOL provides an IDE and unified workflow for electrical, mechanical, fluid, acoustics, and chemical applications. Additionally, simulation results from COMSOL can also be exported to a MATLAB environment to allow for more complicated post-processing.

In this project we use COMSOL Multiphysics 5.5. This software helps model systems by defining the partial chemical equations.

1.3 MONOLITHIC REACTOR

Monoliths are structures that contain various types of interconnected or separated channels (straight, wavy, or crimped) in a single block of material. Monolithic reactors are those filled with monoliths that are either made of porous catalytic material or the catalytic material is deposited (wash coated) in the channels of an inert monolithic support. Monolithic reactor was first used for car-exhaust purification, where the low pressure drop of the monolith compared to packed catalyst bed is a great advantage. The channels normally have circular, square, or triangular cross sections. In our project, channel walls function as catalyst and the channels provide space for flow of gas (Llorca et al. (2015))

2. LITERATURE SURVEY, AIM AND OBJECTIVES

Monolithic structures pose an alternative attractive to conventionally prepared catalyst pellets or powders due to a number of superior properties. They offer better mass transfer, a low pressure drop, thermal stability and good mechanical strength over conventional catalyst pellets (Friedrich et al.(2017))

No one rate equation is not valid for all platinum monolith catalysts, since the value of the constants will depend on the catalyst composition employed, the washcoat thickness and the channel geometry. For example, Doory (1992) studied CO oxidation in 1 mm square monolith channels and proposed a rate equation of the same general form but with constant values as follows: $A_w = 5.1 \times 10^{12} \text{ mol m}^{-2} \text{ s}^{-1}$, $E_w = 93,366 \text{ J mol}^{-1}$, $AA = 469.3$ and $EA = 838 \text{ J mol}^{-1}$, (Hayes et al.(1994))

When CO concentration is very low, such that the value of unity dominates the denominator of eq.(1) and with excess oxygen, the rate is first order. For larger values of CO concentration (for example, at 4% CO in air), the denominator can have values higher than ten. As CO concentration decreases there can be a dramatic change in the rate of reaction, leading to sudden light-off in the reactor. (Hayes et al.(1994))

At low temperatures, the catalytic reactions are slow. As temperature increases, the rate of reaction increases, following an exponential dependence on temperature.

(Holgren et al.(1998))

While considering monolith catalyst, three factors significantly affect the rates and role of mass transfer. These are (1) the wash coat thickness, (2) the channel radius, including its non-uniformity around the channel, and (3) the angular diffusion in the wash coat caused by variable thickness. (Hayes et al.(2004))

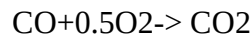
The objective of this thesis is to simulate CO oxidation on monolith catalyst in square and triangular channels.

After simulation, key parameters like inlet temperature is varied from 540-600K to observe changes in conversion and reactor temperature. Feed inlet velocity is varied from 0.15-0.3 m/s to observe changes in conversion

3. METHODOLOGY

3.1 CHEMICAL REACTIONS

CO oxidation to CO₂ can be represented by the following reaction:



The following rate equation for the above equation has been suggested by cavendish (Cavendish et al.(1980))

$$R_{CO} = \frac{k_{CO} C_{CO} C_{O_2}}{(1 + K_{CO} C_{CO})^2} \text{ (mol CO/ cm}^2\text{.Pt.s)} \quad (1)$$

The temperature dependence of the rate and adsorption constant was found to be well approximated by the simple Arrhenius expressions with the constant pre-exponential factors given above.

The concentration of O₂ and CO is low enough that the denominator from the rate equation can be ignored. (Holmgren et al. (1998))

3.2 MODELLING SINGLE CHANNEL

Before analysing monolithic reactor, a model with above reaction is investigated by solving the model with the plug Flow ideal reactor type in Reaction Engineering Module present in COMSOL 5.5

Model Equations

Mass Balance

The mass balance equation for a plug flow reactor under steady condition is given by:

$$\frac{dF_i}{dV} = R_i$$

Where F_i is molar flow rate of specie 'i', V is the reactor volume in V_i , and R_i is the rate of reaction in mol/s m^3

The energy balance for the ideal gas in a reaction is:

$$\sum_i F_i C_{p,i} \frac{dT}{dV} = Q_{ext} + Q$$

Heat due to chemical reaction:

$$Q = \sum_i H_j r_j$$

Where H_j is the heat of the reaction and r_j the reaction rate

we create a linear temperature ramp to investigate impact of inlet temperature. Heat balance equation is used for non isothermal condition which is explained in oxidation in monolithic reactor

Appropriate ratio of CO₂:CO can be determined by analysing the kinetics. Looking at the inlet reaction rate of oxidation as a function of temperature and relative amounts of reactants.

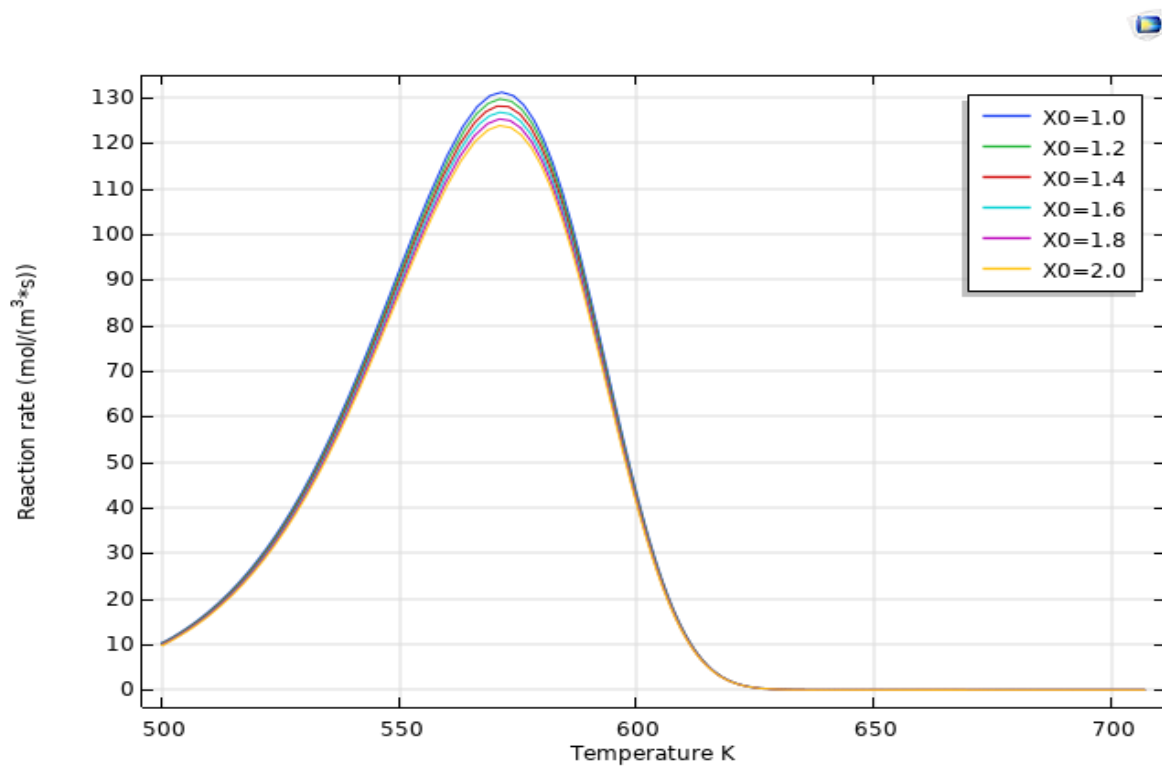


Figure 1: Inlet reaction rates of the CO oxidation reaction as a function of temperature.

The curves represent a set of CO₂:CO ratios ranging from 1 to 2. The concentration of CO in the exhaust gas entering is known to be 1.55E-7 mol/s

The kinetic analysis suggests that preferred working condition involve moderate temperature (~550-600K). The ratio of CO₂:CO doesn't make any significant change in the rate of reaction.

3.3 MODEL

The channels are commonly square or triangular in cross-section, and have a diameter of 0.5-4 mm (Holmgren et al.(1998)).

The 3D Geometry used is taken from COMSOL Application Gallery. The monolithic reactor has modular structure made up of reactive channel blocks and supporting walls. The length of the reactor is 700mm and has a radius of 100mm.

The channel has a cross sectional area of 12.6mm^2 and void fraction of the channel block is 0.75.

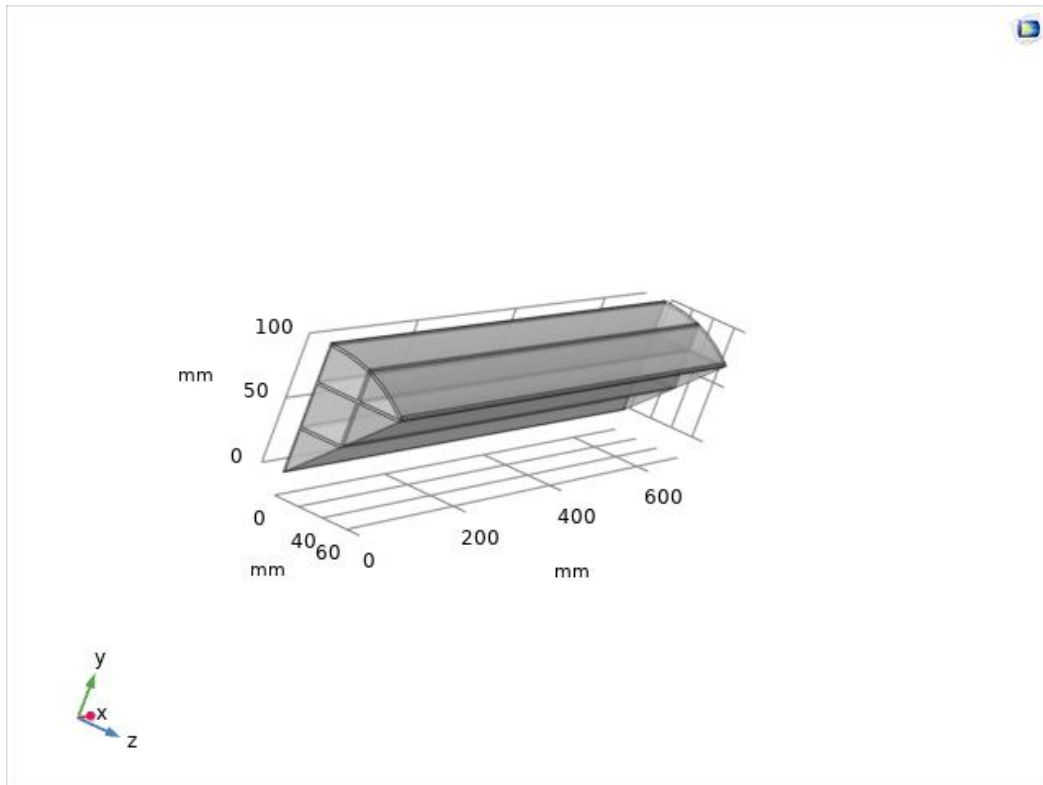


Figure 2: Reaction takes place in the channel blocks. Supporting walls hold together the full reactor geometry. Taking symmetry reduces modelling domain to one-eighth of the full reactor geometry.

3.4 GOVERNING EQUATIONS

Mass is not exchanged between channels so mass transport can be defined by 1D mass transport equation. Assuming fully developed laminar flow is developed in the channel such that average flow field is proportional to pressure difference across the reactor. The fluid flow transports mass and energy only along the channel direction. The Multiphysics problem becomes quite complicated as, each equation shows some temperature dependence.

Beginning with the conservation of mass yields the following equation:

$$\nabla \cdot (-Di \nabla ci) + u \cdot \nabla ci = Ri \quad (1)$$

In Equation 1, Di is the diffusion coefficient, ci is the species concentration, and u is the 3D vector of mass-averaged velocity

Mass transport is considered only along the direction of channels, corresponding to the x-axis in the 3D Geometry used. For diffusive transport this is accomplished by setting remaining components in the diffusivity matrix to zero. The pressure driven convective flow is considered only along x-direction. Each channel behaves thus behaves like a plug flow model with diffusion.

The flow of reactant through monolith is modelled using Darcy's Law with governing equation:

$$\nabla \cdot (\rho u) = 0 \quad (2)$$

The monolith is treated as a porous matrix,

$$u = \frac{\kappa}{\mu} \nabla p \quad (3)$$

The density, ρ and viscosity, μ are assumed to be represented by temperature dependent properties of air

With regards to energy balance in porous monolith reactor describing heat transfer, COMSOL uses the following equation,

$$(\rho C_p)_{eff} \frac{\partial T}{\partial t} + \rho L C_p L u \cdot \nabla T = \nabla (k_{eff} \nabla T) + Q \quad (4)$$

Where ρ_L is the fluid density, C_{pL} is the fluid heat capacity, $(\rho C_p)_{eff}$ is the equivalent volumetric heat capacity, and k_{eff} is equivalent thermal conductivity. u is the average fluid velocity which is calculated in the Darcy's Law interface. Q is the heat source due to exothermic reaction in this case

For steady state, the above equation simplifies to

$$\rho L C_p L u \cdot \nabla T = \nabla (k_{eff} \nabla T) + Q \quad (5)$$

Solid material's volume fraction denoted by θ_p is related to volume fraction of liquid or porosity θ_L by,

$$\theta_p + \theta_L = 1$$

Equivalent conductivity of the solid-fluid system k_{eff} is related to conductivity of solid k_p and to the conductivity of the fluid k_L by

$$k_{eq} = \theta_p k_p + \theta_L k_L \quad (6)$$

Equation 5 is set up for fluid domain. For supporting walls in the reactor, heat transfer occurs by conduction:

$$-\nabla \cdot (k_s \nabla T) = 0 \quad (7)$$

3.5 BOUNDARY CONDITIONS

Species concentration are defined at the reactor inlet boundary:

$$c = c_{in}$$

With regards to outlet, a convective flux condition was set. This convective flux is the mass transfer analogy of equation 1:

$$n.(-D\nabla c) = 0$$

For fluid flow, pressure conditions are set for inlet and outlet boundaries.

For heat transfer, the temperature at the inlet boundaries is:

$$T = T_0$$

At the outlet, use the Outflow condition:

$$n.(k\nabla T) = 0$$

For reactor walls the heat flux through boundaries is given by:

$$q_m.n = h(T - T_{amb})$$

Where h is heat transfer coefficient, and T_{amb} is the ambient temperature.

3.6 THERMODYNAMIC AND TRANSPORT PROPERTIES

The Thermodynamics feature on COMSOL 5.5 provides all necessary properties for the simulation.

Different models are available for calculation of thermal and transport

In the model, nitrogen assumed as solvent. This indicates that beside heat of reaction calculation, other properties are not composition dependent.

3.7 IMPLEMENTED CONDITIONS

Due to various complications resulting from Multiphysics coupling and issues with convergence, the values used in reaction kinetics had to be changed for solution to converge. The kinetic equation used is not valid for all platinum monolith catalysts, since the value of the constants will depend on the catalyst composition employed, the washcoat thickness and the channel geometry (Hayes et al. (1994)).

4. RESULTS

Varying inlet Temperature from 540K to 600K with step size of 20, 3D plots for change in conversion and reactor temperature are obtained

4.1.1 CHANGE IN CONVERSION BY VARYING INLET TEMPERATURE

Keeping other parameters constant; here is the data produced by 3D Model.

T_{in(2)}=540 K Isosurface: ((F_CO_{in}/vrate)-cCO)/(F_CO_{in}/vrate) (1)

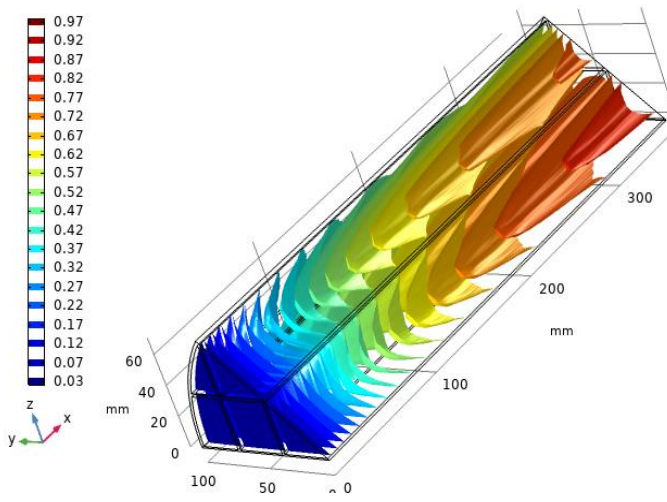


Fig 3.1

T_{in(3)}=560 K Isosurface: ((F_CO_{in}/vrate)-cCO)/(F_CO_{in}/vrate) (1)

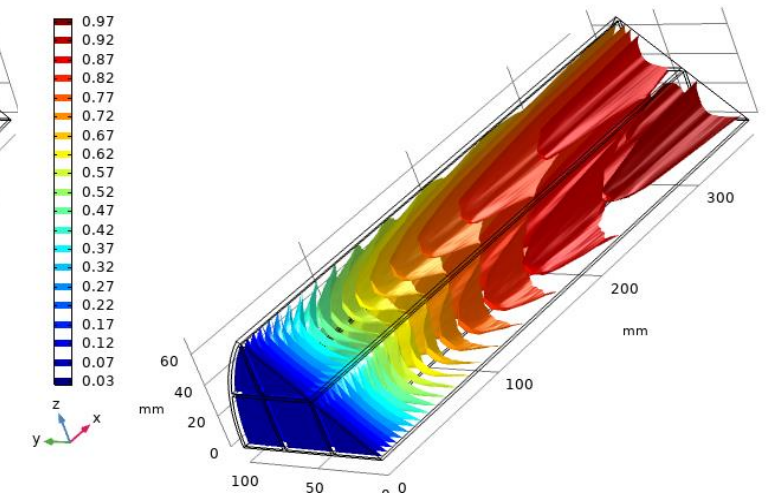


Fig 3.2

T_{in(4)}=580 K Isosurface: ((F_CO_{in}/vrate)-cCO)/(F_CO_{in}/vrate) (1)

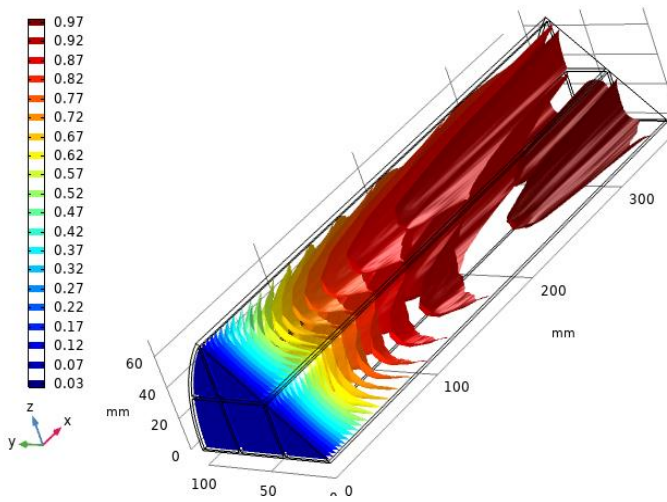


Fig 3.3

T_{in(5)}=600 K Isosurface: ((F_CO_{in}/vrate)-cCO)/(F_CO_{in}/vrate) (1)

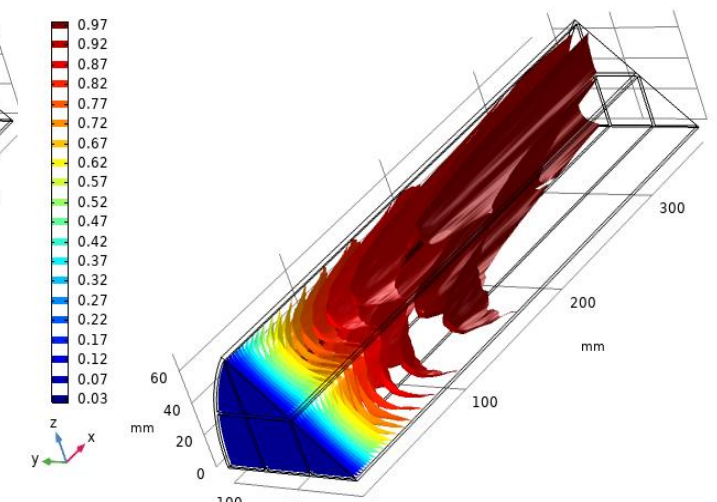


Fig 3.4

Fig 3: Isosurfaces showing the conversion of CO in the reactor for Temperatures 540K,560K,580K and 600K at constant velocity 0.25m/s

With increase in temperature, higher conversion is obtained at any given length. The isosurfaces in the Figure 3 show how the channels performance depends on its position in the reactor, implying that the problem is 3D in nature.

Although individual channels do not exchange mass, they are connected through the temperature distribution in the reactor. The temperature the flow velocity, density of reactants and reaction rates

4.1.2 CHANGE IN CONVERSION BY VARYING AVERAGE INLET VELOCITY

Average inlet velocity is changed, keeping inlet temperature constant at 600K

$v_{av}(1)=0.15$ m/s Isosurface: $((F_{CO_in}/vrate)-cCO)/(F_{CO_in}/vrate)$ (1)

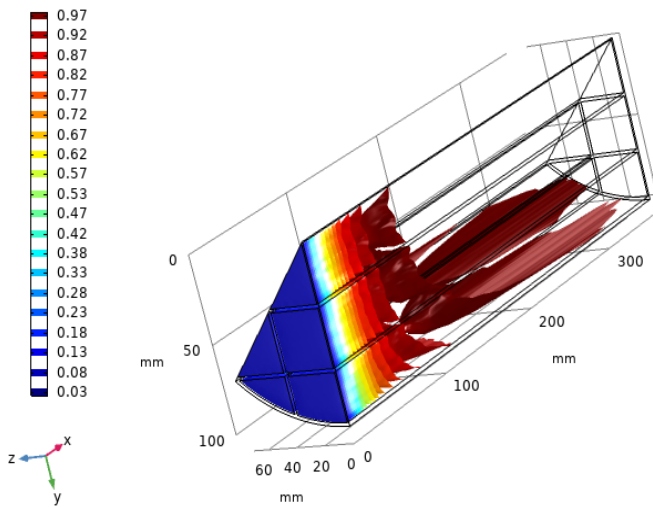


Fig 5.1

$v_{av}(2)=0.2$ m/s Isosurface: $((F_{CO_in}/vrate)-cCO)/(F_{CO_in}/vrate)$ (1)

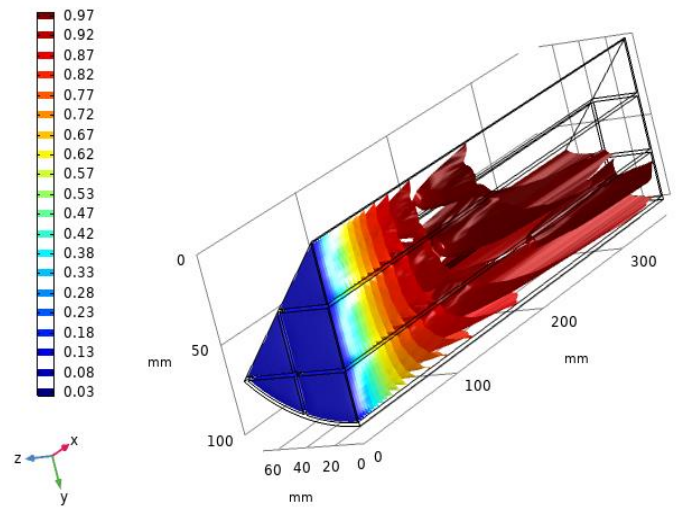


Fig 5.2

$v_{av}(3)=0.25$ m/s Isosurface: $((F_{CO_in}/vrate)-cCO)/(F_{CO_in}/vrate)$ (1)

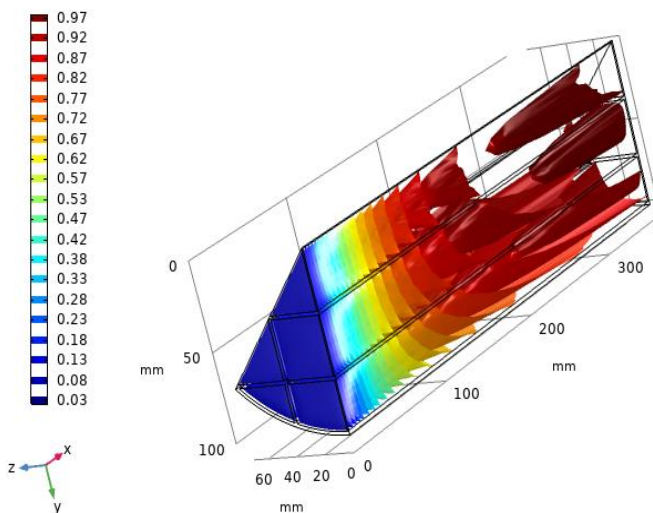


Fig 5.3

$v_{av}(4)=0.3$ m/s Isosurface: $((F_{CO_in}/vrate)-cCO)/(F_{CO_in}/vrate)$ (1)

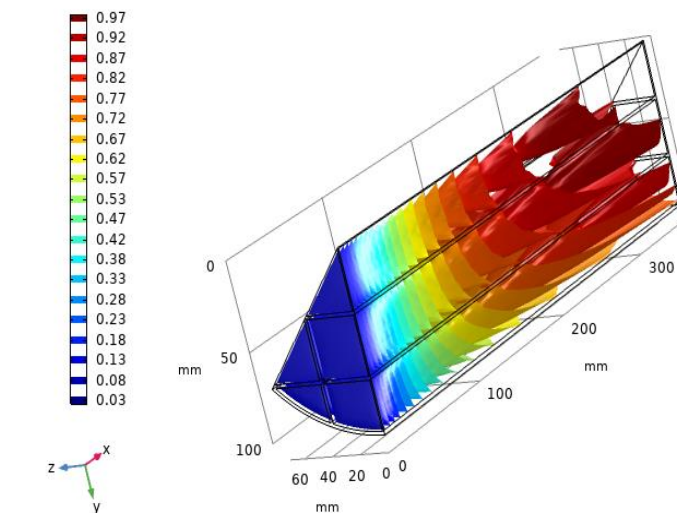


Fig 5.4

Figure 5: conversion in reactor with changing velocity, at temperature=600K

Lower the velocity, lesser is the reactor length required to obtain equivalent conversion

4.2 CHANGE IN REACTOR TEMPERATURE BY VARYING INLET TEMPERATURE

Keeping other parameters constant; Cross sections of the reactor temperature are shown in Figure 4.

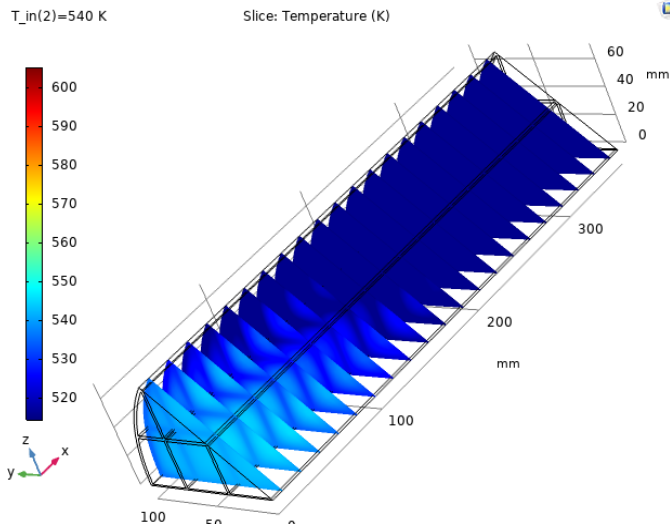


Fig 4.1

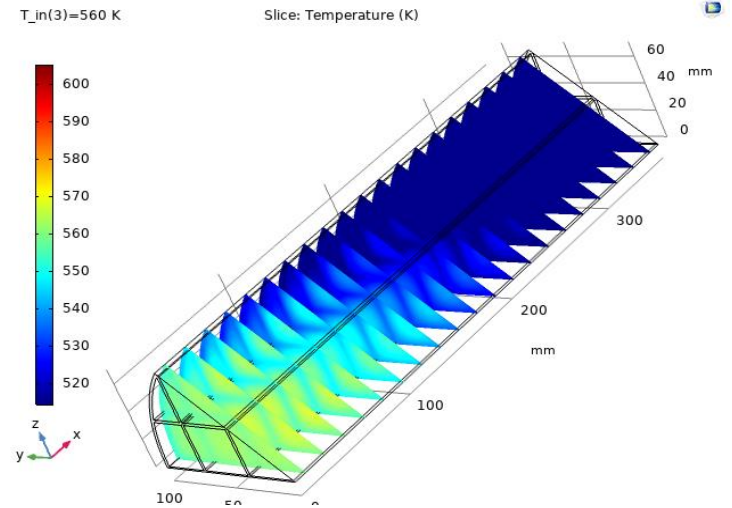


Fig 4.2

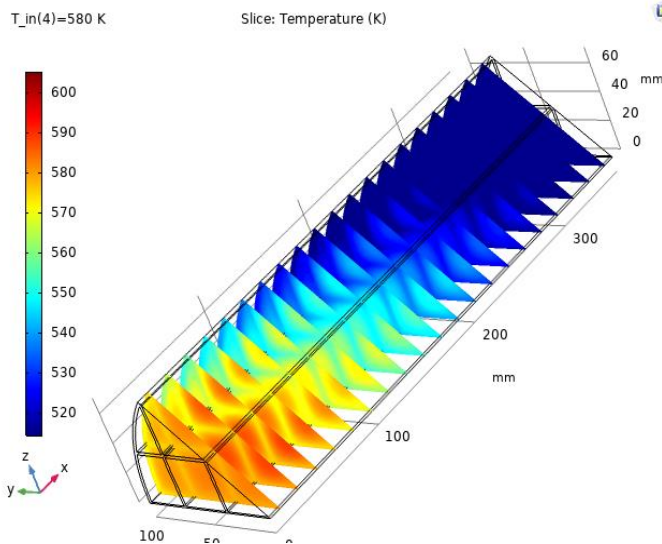


Fig 4.3

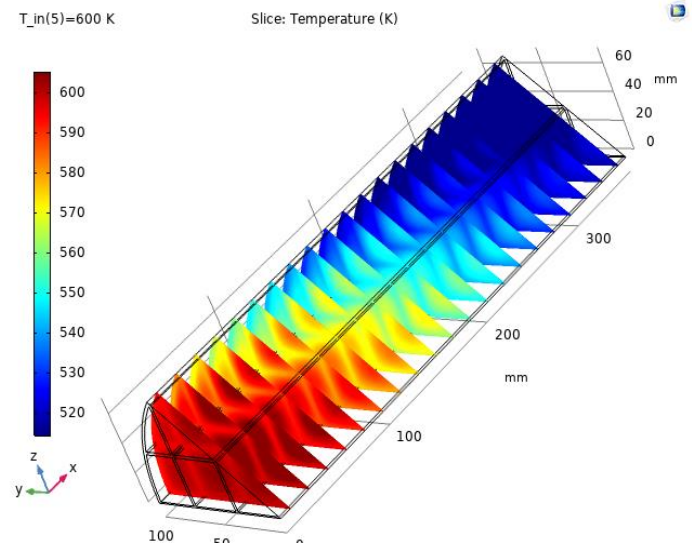


Fig 4.4

Figure 4: Temperature distribution in the cross section of reactor for inlet temperatures 540K, 560K, 580K and 600K and constant velocity of .25m/s

The temperature in the central parts of the reactor increases due to the reaction being exothermic in nature, while the temperature decreases through heat loss to surrounding

5. CONCLUSION

Satisfactory results can be obtained but simulation without experimental data makes it hard to verify accuracy and reliability of a model. This model is more detailed than catalytic packed bed reactor where results obtained were in 2D.

Though implications of this simplified model are limited several trends can be noted that are helpful for understanding the coupling of the overall system. The model is heavily temperature dependent and can be seen from results obtained by changing temperature.

Chemical and thermal coupling is more cyclical, changing temperature changes rate of reaction, which in turn produces change in heat of reaction, which in turn produces change in temperature.

Kinetic parameter in catalytic reaction in a monolith depends on several factors like catalyst used, pore size etc and in absence of values specifically needed for this model are not available, similar parameters and data is taken from COMSOL Application Gallery to make this model converge

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